Unit 1: Basic Internet Skills & Hardware Components

1. Understanding Web Browsers

> **Definition: A web browser is an application software used to access, retrieve, and view information (websites, web pages, videos, etc.) on the World Wide Web (WWW).**

The internet is a vast network of connected computers, and the World Wide Web is the collection of web pages stored on these computers. To read these pages, which are written in HTML (HyperText Markup Language), you need a translator. That translator is the Web Browser.

How it works: When a user enters a Web Address (URL) into the browser, it sends a request to the web server using HTTP/HTTPS protocols. The server responds by sending data packets back. The browser's "Rendering Engine" reads the HTML, CSS, and JavaScript code and displays a user-friendly graphical webpage on the screen.

Modern browsers do much more than just render pages. They manage security, store passwords, block malicious scripts, and offer extensions to enhance functionality.

Key Features of a Browser:

- **Address Bar (Omnibox):** Where you type the URL (e.g., www.rgpv.ac.in) or search queries.
- **Tabs:** Allows opening multiple websites in a single window.
- **Bookmarks/Favorites:** Used to save frequently visited websites for quick access.
- **Cache & Cookies:** Cache stores webpage elements (like images) to load pages faster next time. Cookies store user preferences and login sessions.
- **Incognito/Private Mode:** Browsing mode where history, search records, and cookies are not saved locally.

Examples:

Google Chrome (uses Blink engine), Mozilla Firefox (Gecko), Apple Safari (WebKit), Microsoft Edge.

> 👉 **Hinglish Explanation:** Browser ek software app hai jiske bina hum internet nahi chala sakte. Jab hum www.google.com likhte hain, toh browser server se data mangwata hai aur us coding (HTML) ko design mein badal kar hamari screen par dikhata hai. Chrome, Firefox, Safari sab web browsers ke examples hain. Iska Cache feature page ko agli baar jaldi load karne mein madad karta hai.

2. Efficient Use of Search Engines

> **Definition: A search engine is a web-based tool or software program designed to search for specific**

information on the World Wide Web using keywords provided by the user.

With billions of web pages on the internet, finding exact information without a search engine is impossible. Search engines work in three primary steps: **Crawling** (bots/spiders scan the web for new content), **Indexing** (storing and organizing the scanned data in a massive database), and **Ranking** (using complex algorithms to show the most relevant results first when a user searches).
Efficient Searching Techniques: To get precise results in exams or professional work, one must know how to use "Search Operators". Normal searching often yields millions of irrelevant results. Using operators narrows down the search strictly to what is needed.

Important Search Operators for Efficient Use:

- **Exact Match (" "):** Put quotes around words to find exact phrases. *Example: "Artificial Intelligence Notes"* will only show pages with this exact sentence.
- **Exclude Words (-):** Use a minus sign to exclude a word. *Example: Apple -fruit* will search for the Apple tech company, excluding the fruit.
- **Site Specific (site:):** Search within a specific website. *Example: syllabus site:rgpv.ac.in* will only search the RGPV official website.
- **File Type (filetype:):** Search for specific document types like PDF or PPT. *Example: Python programming filetype:pdf*.
- **OR Operator:** Search for multiple terms. *Example: Java OR Python*.

```
HOW SEARCH ENGINES WORK:
[Web Spiders/Bots] ---> Scan the WWW (Crawling)
       |
       v
[Massive Database] ---> Organize the data (Indexing)
       |
       v
[User types Keyword] -> Algorithm matches keyword -> Displays Results (Ranking)
```

👉 **Hinglish Explanation:** Search engine internet ki dictionary hai. Ye bots (spiders) bhej kar saari websites ka data apne paas save kar leta hai (Indexing). Google aur Bing iske examples hain. Efficient search ka matlab hai smart tareeke se dhoondhna. Agar PDF notes chahiye toh hum direct `filetype:pdf` likh sakte hain, aur agar kisi ek website me kuch dhundhna ho toh `site:website.com` likh sakte hain, isse time bachta hai.

3. Digital India Portals & College Portals

Definition: Digital India is a flagship campaign of the Government of India launched to ensure government services are made available to citizens electronically by improving online infrastructure and internet connectivity.

A "Portal" is a specially designed website that brings information from diverse sources in a uniform way. To empower citizens with e-governance, various national, state, and institutional portals have been developed.

National Portals (Central Government):

- **India.gov.in:** The National Portal of India. It provides a single-window access to information and services provided by the

Indian government.

- **DigiLocker:** A secure cloud-based platform for storage, sharing, and verification of documents & certificates (like Aadhard card, marksheets, driving license) directly from issuers.
- **UMANG (Unified Mobile Application for New-age Governance):** An all-in-one app/portal offering access to pan-India e-Gov services from the Central, State, Local Bodies, and Agencies.
- **NSP (National Scholarship Portal):** A one-stop portal for all government scholarships for students.

State Portals (e.g., MPOnline):

Every state has its own digital portal to handle local citizen services. For example, **MPOnline** is the e-governance portal of Madhya Pradesh. It is used for paying utility bills, applying for state-level exams, domicile certificates, and university counseling. Similarly, Rajasthan has *e-Mitra*, and UP has *e-Sanchar*.

College/University Portals (ERP Systems):

Educational institutions use web portals (often called ERP - Enterprise Resource Planning) to manage academic operations.

- **Student Login:** To check attendance, timetable, fee payment status, and download assignment materials.
- **Faculty Login:** To upload marks, study materials, and manage student attendance.
- **Exam Portals:** Portals like RGPV student login where exam forms are filled, admit cards are downloaded, and results are declared.

Advantages of these Portals:

Paperless operations, 24x7 availability of services, transparency, eradication of middlemen (corruption reduction), and massive time saving.

> 👉 **Hinglish Explanation:** Portal ek aisi website hoti hai jaha bahut saari services ek hi jagah milti hain. Digital India ka goal hai sab kuch online karna. National level par 'DigiLocker' (documents save karne ke liye) aur 'UMANG' hain. State level par jaise MP mein 'MPOnline' hai bill bharne aur forms dalne ke liye. College portals se hum apne results, fees aur attendance ghar baithe dekh sakte hain, jisse college ki line me nahi lagna padta.

4. CPU (Central Processing Unit)

> **Definition: The Central Processing Unit (CPU) is the primary component of a computer that acts as its "brain." It is an electronic circuitry that executes instructions comprising a computer program.**

No computer can function without a CPU. Whether it is a smartphone, a laptop, or a supercomputer, the CPU is responsible for handling all instructions it receives from hardware and software. It operates on the fundamental **Fetch-Decode-Execute** cycle.

Main Components of CPU:

1. **ALU (Arithmetic Logic Unit):** This section performs all mathematical calculations (addition, subtraction, multiplication) and logical operations (AND, OR, NOT, comparisons like >, <, =).
2. **CU (Control Unit):** It acts like a traffic police officer. It does not process data itself; instead, it directs the flow of data between the memory, ALU, and input/output devices. It fetches instructions from memory and decodes them.
3. **Registers:** These are extremely fast, tiny, temporary storage areas located directly inside the CPU. They hold data that is immediately required for processing (e.g., Accumulator, Program Counter).

Performance Factors:

- **Clock Speed:** Measured in Gigahertz (GHz). E.g., a 3.0 GHz CPU executes 3 billion cycles per second.
- **Cores:** Modern CPUs have multiple processing units (Cores) within a single chip (Dual-core, Quad-core, Octa-core), allowing them to perform multitasking effortlessly.

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CPU Architecture:

      +-----------------------------------------+
      |       CENTRAL PROCESSING UNIT (CPU)     |
      |                                         |
      |   +-------------+      +------------+   |
      |   |Control Unit |<---->|   A.L.U.   |   |
      |   |    (CU)     |      |            |   |
      |   +-------------+      +------------+   |
      |          ^                   ^          |
      |          |                   |          |
      |          v                   v          |
      |   +---------------------------------+   |
      |   |            REGISTERS            |   |
      |   +---------------------------------+   |
      +-----------------------------------------+
                         ^ |
           Data/Commands | | Results
                         v v
            +---------------------------+
            |      MAIN MEMORY (RAM)    |
            +---------------------------+
```

👉 **Hinglish Explanation:** CPU computer ka dimaag (brain) hai (jaise Intel Core i5 ya AMD Ryzen). Iske teen main hisse hote hain: ALU (jo math aur logic ka kaam karta hai), CU (Control Unit - jo traffic police ki tarah data ko guide karta hai), aur Registers (sabse fast memory jo turant use aane wala data rakhti hai). CPU ki speed Gigahertz (GHz) me naapi jati hai. Jitne zyada Cores honge, computer utna hi accha multitasking karega.

5. Computer Memory (Primary Storage)

Definition: Computer memory is any physical device capable of storing information temporarily or permanently. Primary memory is the main working area of the computer, directly accessible by the CPU.

Computer memory is organized in a hierarchy based on speed, cost, and capacity. The CPU needs blazing-fast memory to feed it data; otherwise, a fast CPU will sit idle waiting for slow hard drives. Primary memory consists of two main types: RAM and ROM.

1. RAM (Random Access Memory):

- **Nature:** It is **Volatile** memory. This means it loses all its stored data the moment the computer's power is turned off.
- **Function:** It is the computer's "workspace." When you open an app or file, it is loaded from the hard drive into the RAM because RAM is much faster. The CPU directly reads/writes data here.

- **Types:** SRAM (Static RAM - used in Cache memory, very fast/expensive) and DRAM (Dynamic RAM - used as main system memory, requires constant refreshing).
- **Example:** 8GB or 16GB DDR4 RAM in laptops.

2. ROM (Read Only Memory):

- **Nature:** It is **Non-Volatile** memory. Data remains intact even without power.
- **Function:** It is generally written once during manufacturing. It stores crucial, unchangeable instructions required to start (boot) the computer, known as the BIOS (Basic Input/Output System) or firmware.
- **Types:** PROM (Programmable ROM), EPROM (Erasable PROM), EEPROM (Electrically Erasable PROM - used in modern flash drives).

Cache Memory:

A very small, ultra-fast memory placed between the CPU and RAM. It stores the most frequently used instructions so the CPU doesn't have to wait for the relatively slower RAM.

> 👉 **Hinglish Explanation:** Memory 2 type ki hoti hai. **RAM** working table ki tarah hai, jab aap game khelte ho toh game RAM me chalta hai. Ye "Volatile" hai, yani light jate hi iska data ud jata hai. **ROM** ek permanent memory hai jisme computer chalu karne ke instructions (BIOS) company wale fit karke dete hain. Ise mita nahi sakte aasaani se, isliye light jane par bhi ye safe rehta hai.

6. Hard Disk Drive (HDD) & Storage Devices

> **Definition: An HDD is a non-volatile, secondary storage device used to store operating systems, programs, and vast amounts of user data permanently using magnetic storage.**

While RAM is fast, it is temporary and expensive. To store movies, photos, and software permanently, we use Secondary Storage Devices. They have huge capacities (terabytes) but are much slower than RAM.

Hard Disk Drive (HDD):

- **Mechanism:** It contains multiple circular metal disks called **Platters** coated with magnetic material. These platters spin at high speeds (e.g., 5400 or 7200 RPM - Revolutions Per Minute). A mechanical **Read/Write Head** moves across the platters to magnetize areas (writing data as 0s and 1s) or detect magnetic fields (reading data).
- **Structure:** Data on platters is organized into concentric circles called *Tracks*, which are further divided into *Sectors*.
- **Advantages:** High storage capacity at a very low cost.
- **Disadvantages:** Contains moving mechanical parts, making it prone to physical damage from drops. Slower data transfer speeds.

Solid State Drive (SSD) - The Modern Alternative:

Unlike HDDs, SSDs have absolutely **no moving parts**. They store data in microchips (NAND flash memory), similar to a giant pen drive.

- **Advantages:** Extremely fast boot times, very high read/write speeds, silent operation, consumes less battery, and highly durable against shocks.
- **Disadvantages:** More expensive per Gigabyte compared to HDDs.

👉 **Hinglish Explanation:** HDD computer ka permanent godown hai jisme saari files save hoti hain (jaise Local Disk C, D). Iske andar CD jaisi platters ghumti hain aur ek suyi (read/write head) data padhti hai. Kyuki isme motor lagi hai, ye dheema hota hai aur girne par kharab ho sakta hai. Aajkal **SSD** aati hai jisme koi motor nahi hoti, wo chip (flash memory) pe kaam karti hai isliye wo HDD se 10-guni fast hoti hai, par mehengi aati hai.

7. Input Devices: Keyboard and Mouse

Definition: Input devices are pieces of hardware used to provide data and control signals to an information processing system (the computer).

Computers cannot understand human intent automatically. We must translate our commands into binary electronic signals. Input devices do this translation.

Keyboard:

The keyboard is the most common text-based input device. Its layout is typically modeled after the traditional typewriter (QWERTY layout).

- **Key Types:** Alphanumeric keys (A-Z, 0-9), Punctuation, Modifier keys (Shift, Ctrl, Alt), Navigation keys (Arrows), and Function keys (F1 to F12).
- **Working:** Beneath the keys is a circuit board matrix. When a key is pressed, it completes a circuit, and a microprocessor inside the keyboard sends a specific scan code to the CPU, telling it which key was pressed.
- **Types:** Membrane keyboards (rubber dome, common in laptops) and Mechanical keyboards (individual physical switches, preferred by gamers/typists).

Mouse:

A mouse is a hand-held **Pointing Device** that detects two-dimensional motion relative to a surface. It translates this motion into the movement of a pointer (cursor) on the display.

- **Working:** Early mice used a rolling rubber ball. Modern mice use optical sensors (LEDs) or lasers to track movement across a desk by taking thousands of microscopic pictures per second and comparing them.
- **Actions:** Left-click (select/open), Right-click (context menu), Scroll wheel (navigating pages up/down), Drag and drop.
- **Types:** Wired (USB), Wireless (Bluetooth or 2.4GHz Dongle), Trackpad (in laptops).

👉 **Hinglish Explanation:** Input devices se hum computer ko order dete hain. **Keyboard** text type karne ke kaam aata hai. Isme alag-alag keys hoti hain (F1-F12, Alphabets, numbers). **Mouse** ek pointing device hai jisse hum screen par arrow (cursor) chalate hain. Pehle mouse me rubber ki ball hoti thi, aajkal wale **Optical Mouse** me neeche LED light aur sensor laga hota hai jo movement track karta hai.

8. Output Devices: Display / Monitor

Definition: Output devices translate the processed machine-readable data (binary) from the computer into a human-readable form. The Display (Monitor) is the primary visual output device.

The CPU processes data, and the Graphics Processing Unit (GPU) renders it into images. The monitor is the canvas where these images are displayed.

Key Terminology:

- **Pixel (Picture Element):** The smallest single dot of color on a screen. Millions of pixels combine to form an image.
- **Resolution:** The total number of pixels displayed, expressed as Width x Height (e.g., 1920x1080 is Full HD, 3840x2160 is 4K). Higher resolution means sharper, clearer text and images.
- **Refresh Rate:** Measured in Hertz (Hz). It is the number of times the screen updates its image per second. A standard monitor is 60Hz; gaming monitors are 144Hz or higher for smoother motion.

Types of Display Technologies:

- **CRT (Cathode Ray Tube):** The old, bulky, heavy monitors resembling vintage TVs. They used electron guns hitting a phosphorus screen. Obsolete now.
- **LCD (Liquid Crystal Display):** Uses liquid crystals sandwiched between polarized glass. Requires a backlight (usually CCFL tubes) to make the image visible.
- **LED (Light Emitting Diode):** Technically an LCD, but it uses highly efficient LEDs for backlighting instead of bulky tubes. This makes them thinner, brighter, and energy-efficient. (Most common today).
- **OLED (Organic LED):** The most advanced. Each pixel produces its own light. When a pixel shows black, it completely turns off, resulting in infinite contrast, true blacks, and brilliant colors.

> 👉 **Hinglish Explanation:** Output devices result dikhate hain. **Monitor** TV jaisa hota hai. Screen chhote-chote dots se bani hoti hai jinhe **Pixels** kehte hain. Jitne zyada pixels (Resolution), image utni clear hogi (jaise 1080p ya 4K). Pehle ke dabbo wale monitor CRT hote the. Aajkal **LED** chalte hain jo patle hote hain aur power kam khate hain. Sabse premium **OLED** hota hai jisme har pixel ki apni light hoti hai.

9. Other Peripheral Devices

> **Definition: Peripheral devices are auxiliary (extra) hardware components attached to the computer system to expand its functionality. They are not part of the core computing process (like CPU/RAM).**

Peripherals can be input, output, or storage devices. They usually connect externally via USB, HDMI, Bluetooth, or audio jacks. Modern peripherals are "Plug-and-Play," meaning the OS automatically recognizes them without complex setups.

Common Peripheral Devices:

- **Printers (Output):** Produce a hard copy (physical paper) of electronic documents.
  - *Inkjet Printers:* Spray tiny microscopic droplets of liquid ink onto paper. Great for high-quality color photos.
  - *Laser Printers:* Use a laser beam to heat powdered ink (Toner) onto paper. Extremely fast and cost-effective for large volumes of black-and-white text.
- **Scanners (Input):** Optically scan physical images, printed text, or objects and convert them into digital images (soft copy). Often combined with printers today.
- **Speakers & Headphones (Output):** Convert digital audio signals generated by the computer's sound card into analog sound waves (acoustic energy) that humans can hear.
- **Webcam (Input):** A digital video camera that captures moving images and audio in real-time, used for video conferencing

(Zoom/Skype) and live streaming.

- **UPS (Uninterruptible Power Supply):** Provides emergency power from an internal battery when the main power source fails, protecting hardware from sudden shutdowns.

👉 **Hinglish Explanation:** Peripheral devices wo extra cheezein hain jo hum computer me bahar se lagate hain (USB se) taaki computer aur kaam kar sake. **Printer** se hum document ka printout (Hard copy) nikalte hain. **Scanner** paper ko scan karke computer me (Soft copy) daalta hai. Speakers, Webcam, Pen Drives—ye sab peripheral devices hi kahlati hain. Agar CPU me speaker na lagao, toh bhi CPU chalega, isliye inhe auxiliary (extra) device kehte hain.

🔁 Quick Revision Notes (Exam Saver)

- **Browsers vs Search Engines:** Browser is the App (Chrome) to open sites. Search Engine is the tool (Google) to find sites.
- **Portals:** DigiLocker (documents), UMANG (all-in-one e-Gov), MPOnline (state services).
- **Search Operators:** "Exact phrase", -exclude, site:website.com, filetype:pdf.
- **CPU:** The Brain. Contains ALU (calculations), CU (controls flow), Registers (fastest memory).
- **RAM vs ROM:** RAM = Volatile (temporary workspace). ROM = Non-Volatile (permanent BIOS).
- **HDD vs SSD:** HDD uses spinning magnetic disks (slow, cheap). SSD uses flash chips (fast, no moving parts).
- **I/O Basics:** Keyboard & Mouse = Input. Monitor & Printer = Output.
- **Display:** Pixels make up resolution. LCD/LED use backlight; OLED pixels light themselves.

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